

M3 - Discovering Latent Musical Structure Using Spotify Audio Features

Course: CS 4412 – Data Mining
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Implementation Status

The analysis pipeline is now fully functional. From data intake to final interpretation, the entire process is functional and yields significant outcomes.

Nearly 114,000 Spotify songs with 21 properties make up the dataset. These features include categorical identifiers like artist and genre as well as numerical characteristics like danceability, energy, loudness, acousticness, and tempo. The dataset is clean, with few missing values and no duplicate entries, confirming its appropriateness for analysis, according to the first preprocessing procedures.

The pipeline includes the following completed components:

- Data loading and inspection
- Data quality and assessment (missing values and duplicates)
- Feature selection of relevant numerical attributes
- Feature scaling using standardization
- Dimensionality reduction using Principal Component Analysis (PCA)
- Clustering using K-Means
- Density based clustering using DBSCAN
- Anomaly detection using Local Outlier Factor (LOF)
- Evaluation of clustering performance using multiple metrics
- Visualization of results (histograms, heatmaps, PCA projections, clustering plots)

Every strategy intended for this milestone has been successfully carried out. The notebook produces repeatable results and operates flawlessly from start to finish. The analysis now offers significant insights into the dataset, beyond the exploratory phases.

Analytical Choices

The main discovery question served as the basis for the analytical strategy:
What unique musical patterns appear in Spotify audio characteristics, and can unsupervised learning methods reveal significant latent structures?

Several complementary methods were chosen to investigate the dataset from various angles in order to address this.

Because of its effectiveness and capacity to detect global structure in numerical datasets, K-Means clustering was selected as a baseline technique. It is useful for finding broad categories of songs based on their aural characteristics because it clearly divides data into clusters.

In order to investigate whether the data contains irregularly shaped clusters or density-based structures that K-Means might miss, DBSCAN was introduced as an alternate clustering technique. DBSCAN is helpful for identifying non-uniform patterns in the data because, in contrast to K-Means, it does not assume spherical clusters and can detect noise points.

To identify anomalies, the Local Outlier Factor (LOF) was used. By identifying points that differ from their local neighborhood density, this technique makes it possible to find songs with peculiar feature combinations. By emphasizing fringe instances that might not fit into any dominating group, LOF enhances clustering.

For both dimensionality reduction and visualization, Principal Component Analysis (PCA) was employed. Because the dataset is high-dimensional, PCA makes it possible to visualize the underlying structure in two dimensions and lessens redundancy among connected features. Additionally, it enhances the interpretability of clustering results and computational efficiency.

When combined, these techniques offer a thorough analytical framework. A more comprehensive comprehension of the data is made possible by combining clustering, density-based approaches, and anomaly detection rather than depending just on one method.

Results and Interpretation

The application of PCA revealed that the first two principal components capture approximately 40% of the total variance. This indicates that while PCA reveals some underlying structure, a significant portion of variability remains distributed across higher dimensions. This suggests that musical characteristics are complex and influenced by multiple interacting features rather than a small number of dominant factors. K-Means clustering identified three primary clusters within the dataset. Analysis of cluster centroids revealed distinct musical profiles:

- Cluster 0: High energy and loudness, representing energetic or upbeat tracks.
- Cluster 1: Higher danceability and valence, indicating rhythmic and positive songs.
- Cluster 2: High acousticalness and lower energy, representing calmer or acoustic tracks.

These clusters reflect meaningful distinctions in musical style and confirm that audio features can capture recognizable patterns in music. However, evaluation metrics indicate moderate clustering performance. The silhouette score (~ 0.136) suggests weak separation between clusters, while the Davies-Bouldin score indicates overlap between cluster boundaries. This implies that although clusters exist, they are not sharply defined. Instead, songs appear to transition gradually across styles.

DBSCAN clustering produced a large number of small clusters along with many noise points. This indicates that the dataset lacks strong density-based groupings and does not naturally form dense, well-separated regions. This result suggests that music data is more continuously distributed rather than organized into compact clusters. LOF anomaly detection identified a small number of outliers. These points represent songs with unusual combinations of features that differ from typical patterns. These anomalies may correspond to niche, experimental, or genre-blending tracks that do not fit standard musical categories.

Overall, the results demonstrate that while patterns exist in the data, musical structure is inherently complex and continuous rather than strictly segmented.

These findings highlight the complexity of musical structure and demonstrate that while machine learning can identify patterns, these patterns reflect continuous variation rather than strict categorical boundaries.

Discovery Question Progress

The analysis provides meaningful progress toward answering the central discovery question. Unsupervised learning techniques were able to uncover latent structure within the dataset. K-Means clustering revealed broad groupings of songs based on energy, acousticness, and rhythmic characteristics. These groupings align with intuitive categories of music, such as energetic, danceable, and acoustic styles.

However, the findings also show that these structures are not sharply defined. The overlap between clusters and relatively low silhouette score indicate that musical characteristics exist on a spectrum rather than within discrete categories. This reflects the real-world nature of music, where genres and styles often blend and evolve.

DBSCAN results further reinforce this conclusion by showing the absence of strong density-based clusters. Instead of forming compact groups, the data appears to be more uniformly distributed across feature space. Anomaly detection adds another layer of insight by identifying songs that deviate from typical patterns. These outliers highlight the diversity of musical expression and suggest the presence of unique or unconventional tracks.

Overall, the discovery question can be partially answered: Unsupervised techniques can reveal meaningful patterns and groupings, but these patterns reflect gradual transitions rather than rigid boundaries between musical styles.

M4 Plan

While the implementation is complete, several improvements are planned for the final milestone (M4) to enhance the quality and depth of the analysis.

First, the interpretation of results will be further refined to provide deeper domain insights and stronger connections to real-world musical behavior. This includes expanding explanations of cluster characteristics and anomaly significance.

Second, additional techniques may be explored to improve interpretability. For example, classification methods such as decision trees could be used to explain what features distinguish clusters, providing more interpretable rules.

Third, in order to enhance model performance, parameter adjustment will be carried out. To better capture possible structure in the data, this involves experimenting with alternative values of k in K-Means and modifying DBSCAN parameters (epsilon and minimum samples).

Fourth, the quality of visualizations will be enhanced for use in presentations. Clearer labeling, more refined plots, and possibly more visualizations to better convey findings are some examples of this.

In order to produce a coherent and expert final report, the overall story and presentation will be polished. This entails arranging ideas into a coherent narrative and resolving any analysis's lingering shortcomings.